LAB Assignment No 2

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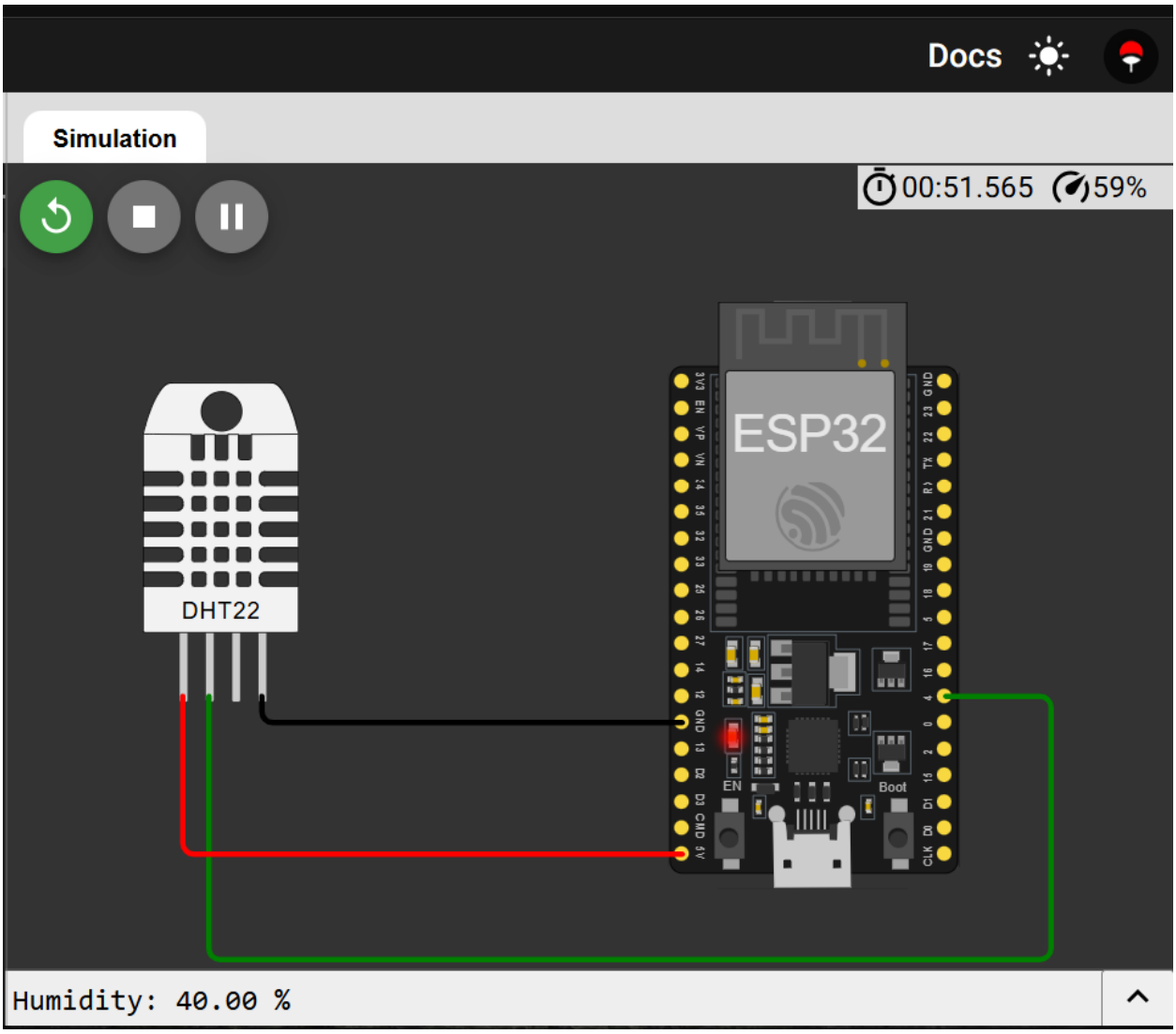
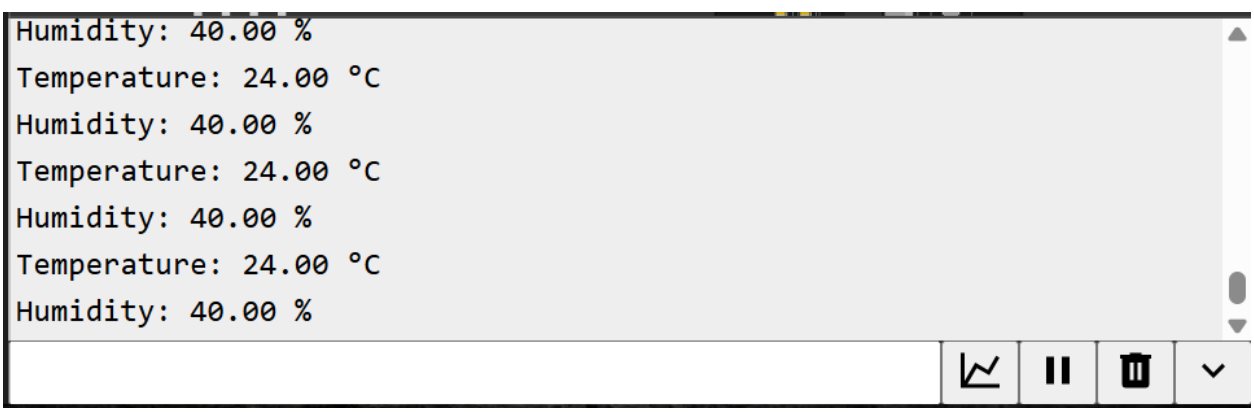
Exercise No. 1 (Interfacing with ESP32: Temp and humidity sensor)

Code:

```
#include "DHT.h"
#define DHTPIN 4
#define DHTTYPE DHT22

DHT dht(DHTPIN, DHTTYPE);

void setup() {
  Serial.begin(115200);
  dht.begin();
}

void loop() {

  float humidity = dht.readHumidity();
  float temperature = dht.readTemperature();

  Serial.print("Temperature: ");
  Serial.print(temperature);
  Serial.println(" °C");

  Serial.print("Humidity: ");
  Serial.print(humidity);
  Serial.println(" %");

  delay(2000);
```

Simulation Diagram:





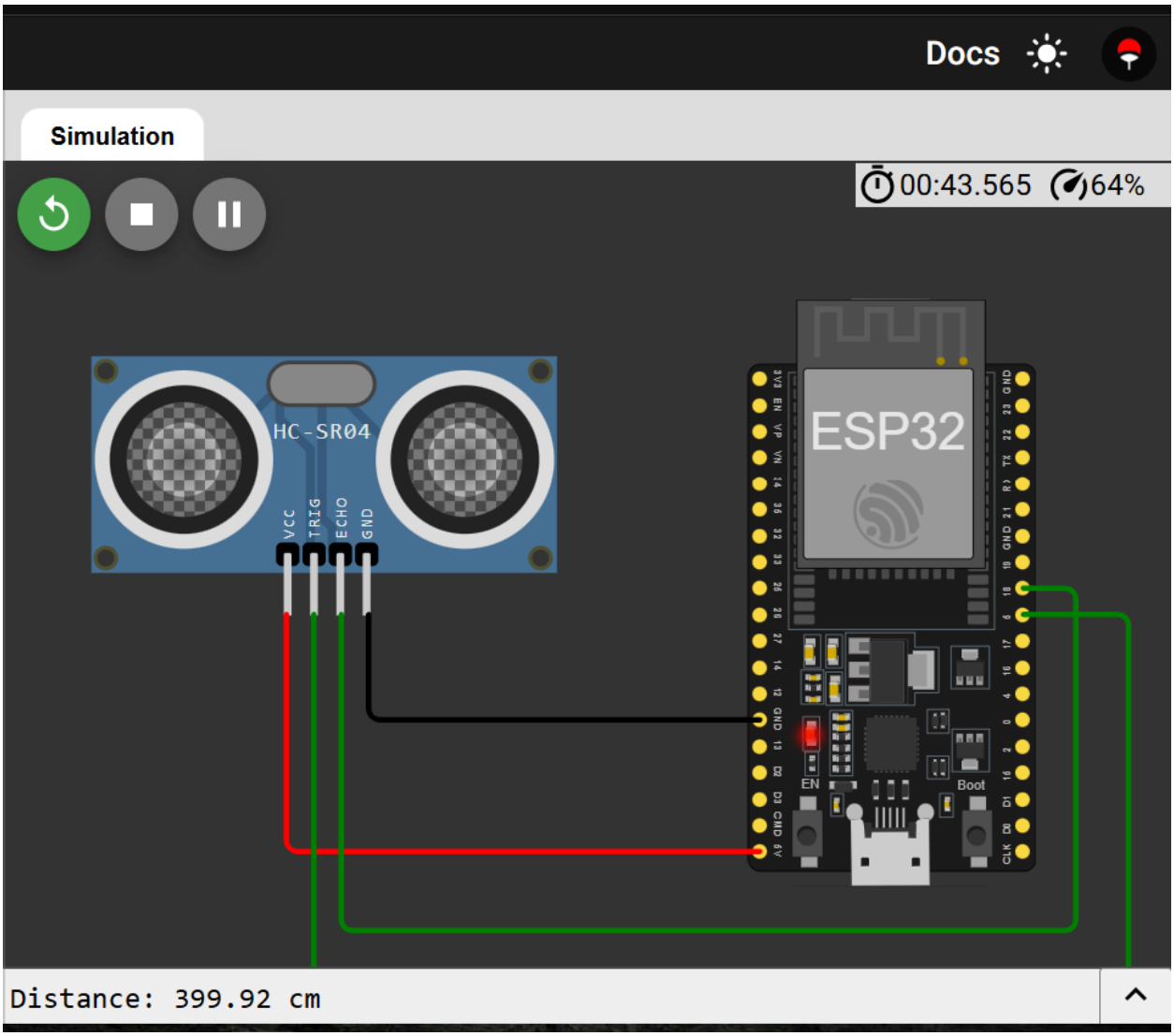
Exercise No. 2 (Interfacing with ESP32: Ultrasonic Sensor)

Code:

```
#define trigPin 5
#define echoPin 18

void setup() {
  Serial.begin(115200);
  pinMode(trigPin, OUTPUT);
  pinMode(echoPin, INPUT);
}

void loop() {

  long duration;
  float distance;

  digitalWrite(trigPin, LOW);
  delayMicroseconds(2);

  digitalWrite(trigPin, HIGH);
  delayMicroseconds(10);
  digitalWrite(trigPin, LOW);

  duration = pulseIn(echoPin, HIGH);

  distance = duration * 0.034 / 2;
```

```
  Serial.print("Distance: ");
  Serial.print(distance);
  Serial.println(" cm");

  delay(1000);
}
```

Simulation Diagram:

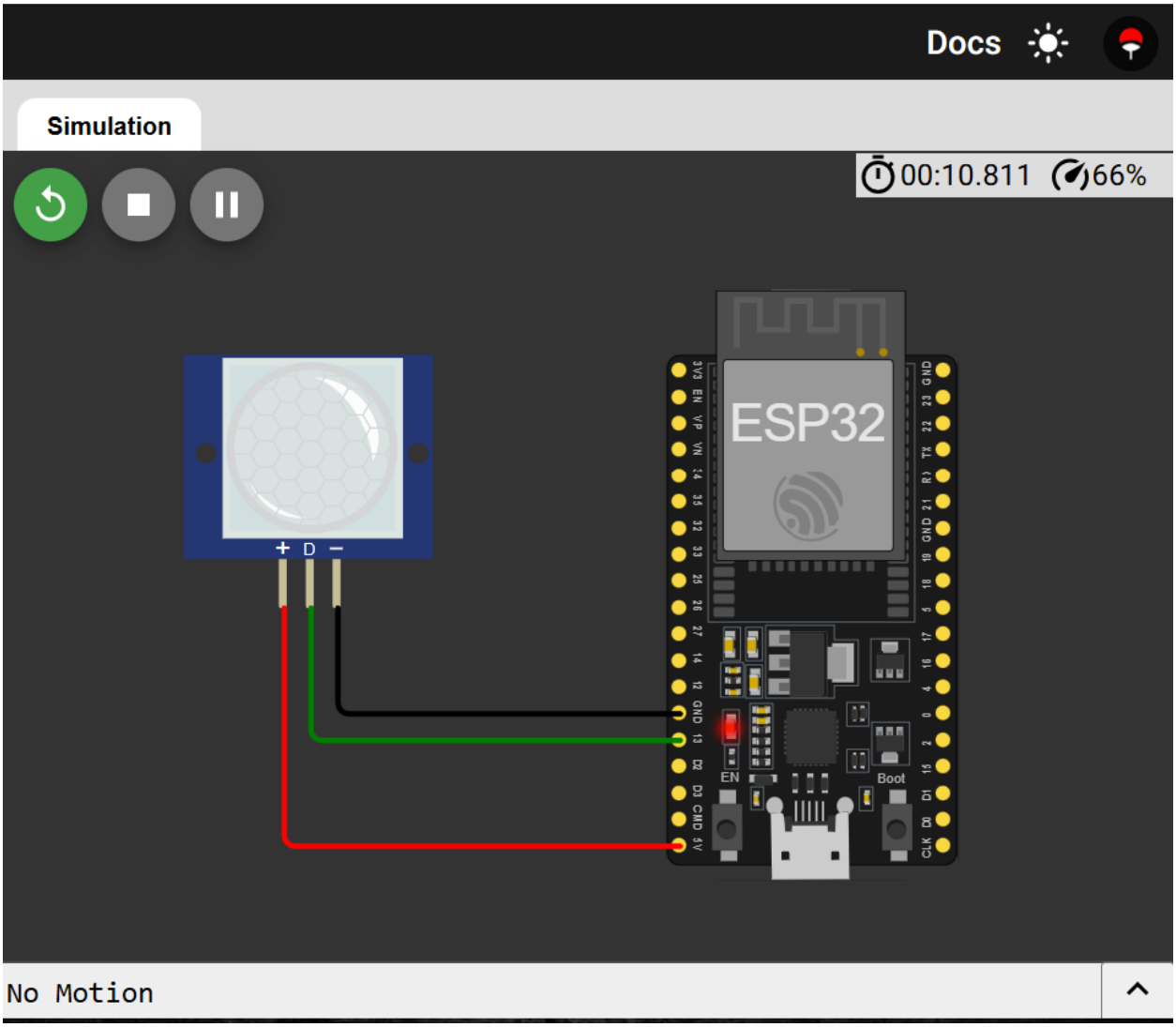


Exercise No. 3 (Interfacing with ESP32: PIR sensor)

Code:

```
#define PIR_PIN 13

void setup() {
  Serial.begin(115200);
  pinMode(PIR_PIN, INPUT);
}

void loop() {

  int motion = digitalRead(PIR_PIN);

  if(motion == HIGH)
    Serial.println("Motion Detected");
  else
    Serial.println("No Motion");

  delay(1000);
}
```

Simulation Diagram:




```
No Motion
No Motion
No Motion
No Motion
No Motion
No Motion
No Motion
```

LAB Review Questions

Question 1: DHT11 with ESP32 using I2C LCD

Interface the **DHT11 Temperature and Humidity Sensor** with **ESP32** and display the temperature and humidity values on an **I2C LCD (16x2)**.

Tasks:

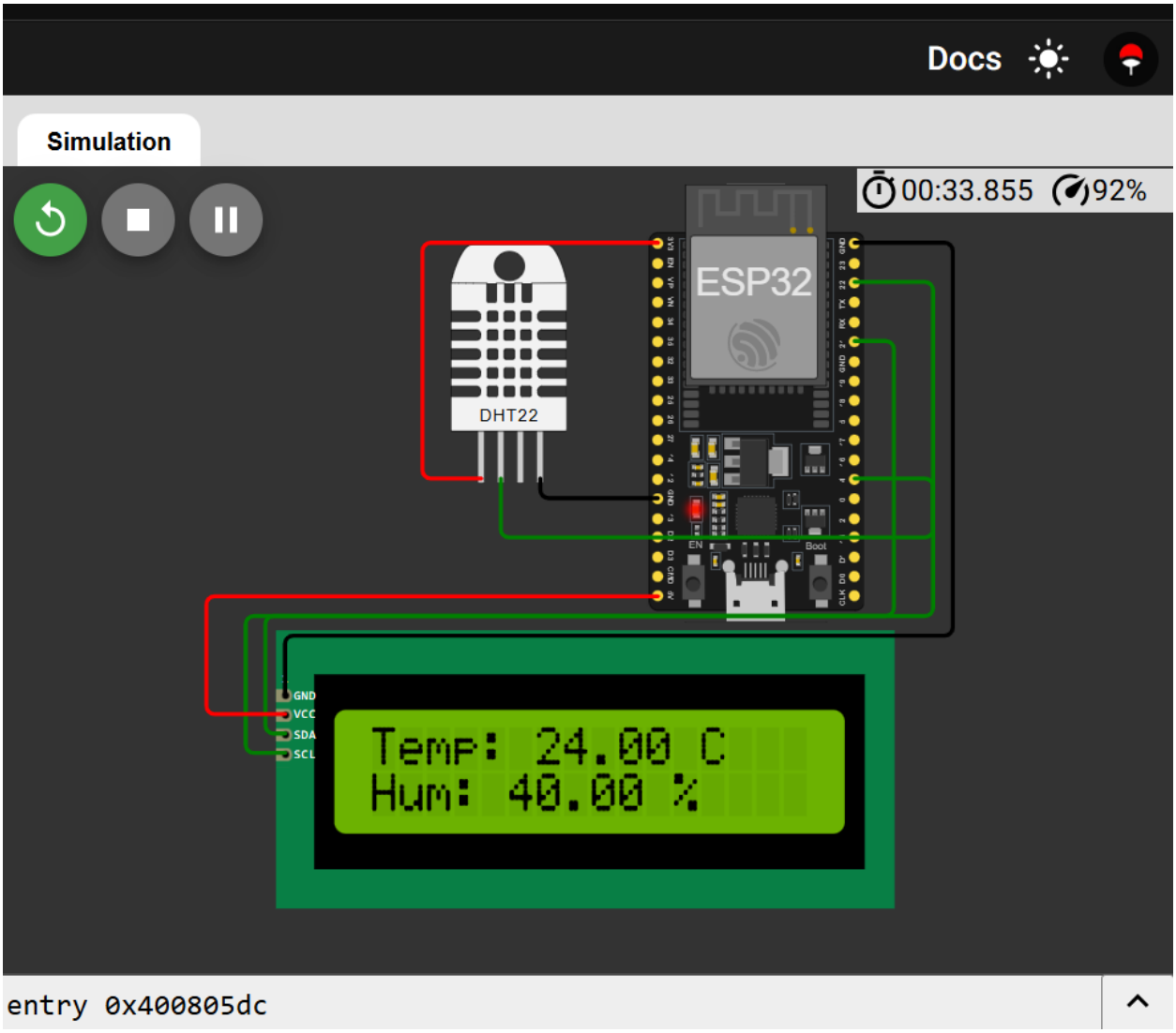
1. Write the **ESP32 program code** to read temperature and humidity from the DHT11 sensor.
2. Display the **temperature and humidity values on the I2C LCD**.
3. Draw the **complete circuit diagram / wiring diagram** of ESP32, DHT11, and I2C LCD.
4. Perform the **simulation using Wokwi or any online simulator**.
5. Capture and attach **screenshots of the running simulation and output**.

CODE:

```
#include <Wire.h>
#include <LiquidCrystal_I2C.h>
#include <DHT.h>

#define DHTPIN 4
#define DHTTYPE DHT22

DHT dht(DHTPIN, DHTTYPE);

// LCD address usually 0x27
LiquidCrystal_I2C lcd(0x27, 16, 2);

void setup() {

  Serial.begin(115200);

  dht.begin();

  lcd.init();
  lcd.backlight();

  lcd.setCursor(0,0);
  lcd.print("DHT11 Sensor");
  delay(2000);
  lcd.clear();
```

```
}

void loop() {

  float humidity = dht.readHumidity();
  float temperature = dht.readTemperature();

  lcd.setCursor(0,0);
  lcd.print("Temp: ");
  lcd.print(temperature);
  lcd.print(" C");

  lcd.setCursor(0,1);
  lcd.print("Hum: ");
  lcd.print(humidity);
  lcd.print(" %");

  delay(2000);
}
```

Simulation Diagram:



Question 2: Display Data on I2C LCD using ESP32

Create a project where **ESP32 displays data on an I2C LCD**.

Tasks:

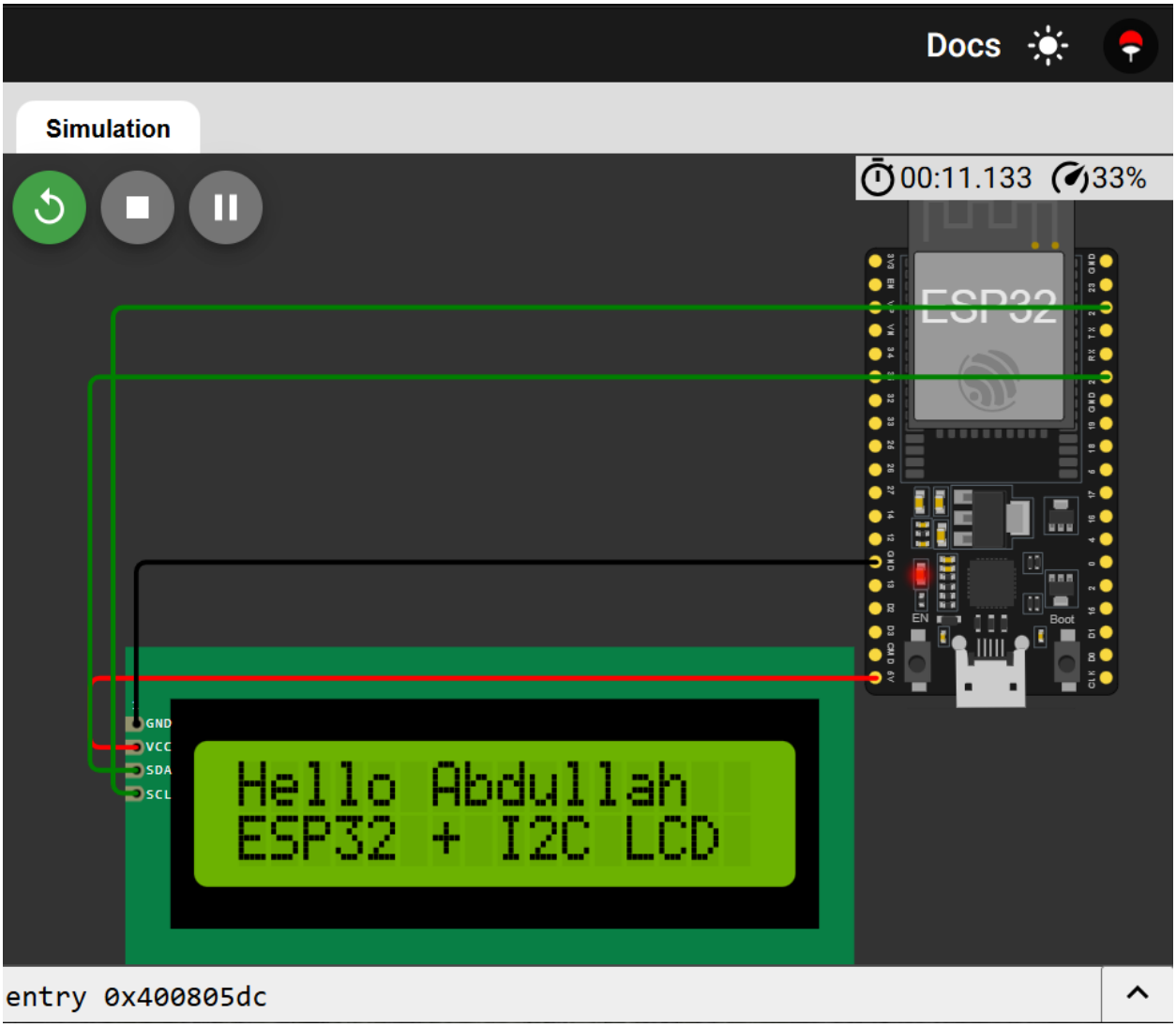
1. Interface **I2C LCD (16x2)** with ESP32.
2. Write a program
3. Draw the **pin connection diagram** between ESP32 and I2C LCD.
4. Run the project in **Wokwi simulation** and attach the screenshot of the output.

CODE:

sketch.ino ● diagram.json ● Library Manager libraries.txt

```
#include <Wire.h>
#include <LiquidCrystal_I2C.h>

LiquidCrystal_I2C lcd(0x27, 16, 2);

void setup() {

  lcd.init();
  lcd.backlight();

  lcd.setCursor(0,0);
  lcd.print("Hello Abdullah");

  lcd.setCursor(0,1);
  lcd.print("ESP32 + I2C LCD");
}

void loop() {

}
```

Simulation Diagram:



Question 3: PIR Motion Sensor with ESP32 and I2C LCD

Interface a **PIR Motion Sensor** with **ESP32** and display the motion detection status on an **I2C LCD**.

Tasks:

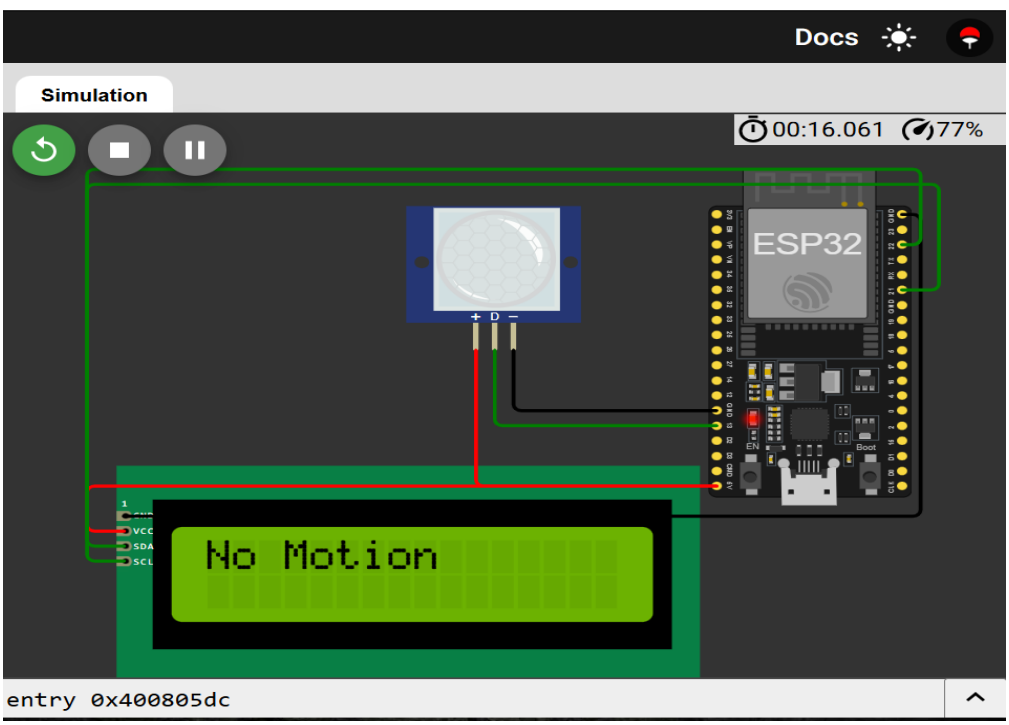
1. Write ESP32 code to read the **digital output of the PIR sensor**.
2. Display the following messages on the LCD:
    - **"Motion Detected"** when motion is detected.

- **“No Motion”** when no motion is detected.

3. Draw the **complete wiring diagram** of ESP32, PIR sensor, and I2C LCD.
4. Perform **simulation in Wokwi** and attach screenshots of the results.

CODE:

```
#include <Wire.h>
#include <LiquidCrystal_I2C.h>

#define PIR_PIN 13

LiquidCrystal_I2C lcd(0x27, 16, 2);

void setup() {

  pinMode(PIR_PIN, INPUT);

  lcd.init();
  lcd.backlight();

  lcd.setCursor(0,0);
  lcd.print("PIR Sensor Ready");
  delay(2000);
  lcd.clear();
}

void loop() {

  int motion = digitalRead(PIR_PIN);

  if (motion == HIGH) {
```

```
    lcd.clear();
    lcd.setCursor(0,0);
    lcd.print("Motion Detected");

  }
  else {

    lcd.clear();
    lcd.setCursor(0,0);
    lcd.print("No Motion");

  }

  delay(1000);
}
```

Simulation Diagram: